

HW11

代靖涵 25120222201319

Exercise 11.1

设随机变量 X 和 Y 独立同分布, 且

$$P(X = -1) = P(Y = -1) = P(X = 1) = P(Y = 1) = \frac{1}{2}.$$

试求 $P(X = Y)$.

Solution.

$$\begin{aligned} P(X = Y) &= P(X = Y = -1) + P(X = Y = 1) \\ &= P(X = -1)P(Y = -1) + P(X = 1)P(Y = 1) \\ &= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

◇

Exercise 11.2

甲、乙两人独立地各进行两次射击, 假设甲的命中率为 0.2, 乙的命中率为 0.5, 以 X 和 Y 分别表示甲和乙的命中次数, 试求 $P(X \leq Y)$.

Proof.

$$P(X = 0) = 0.64, \quad P(X = 1) = 0.32, \quad P(X = 2) = 0.04$$

$$P(Y = 0) = 0.25, \quad P(Y = 1) = 0.5, \quad P(Y = 2) = 0.25$$

$$\begin{aligned} P(X \leq Y) &= P(X = 0)P(Y \geq 0) + P(X = 1)P(Y \geq 1) + P(X = 2)P(Y = 2) \\ &= 0.64 \times 1 + 0.32 \times 0.75 + 0.04 \times 0.25 \\ &= 0.89 \end{aligned}$$

□

Exercise 11.3

设随机变量 X 与 Y 相互独立, 其联合分布列为

X	Y		
	y_1	y_2	y_3
x_1	a	$1/9$	c
x_2	$1/9$	b	$1/3$

试求联合分布列中的 a, b, c .

Proof. 根据图表

$$P(Y = y_3) = c + \frac{1}{3}$$

$$P(Y = y_2) = \frac{1}{9} + b$$

$$P(Y = y_1) = a + \frac{1}{9}$$

由独立性

$$P(X = x_1) = \frac{P(X = x_1, Y = y_1)}{P(Y = y_1)} = \frac{a}{a + \frac{1}{9}} = \frac{9a}{9a + 1}$$

$$P(X = x_2) = \frac{P(Y = y_1, X = x_2)}{P(Y = y_1)} = \frac{\frac{1}{9}}{a + \frac{1}{9}} = \frac{1}{9a + 1}$$

$$c = P(X = x_1)P(Y = y_3) = \frac{9a}{9a + 1} \left(c + \frac{1}{3} \right) \implies \frac{3c}{3c + 1} = \frac{9a}{9a + 1}$$

从而

$$\frac{1}{3c + 1} = \frac{1}{9a + 1}$$

得到

$$c = 3a$$

$$b = P(X = x_2)P(Y = y_2) = \frac{1}{9a + 1} \left(\frac{1}{9} + b \right) \implies \frac{9b}{9b + 1} = \frac{1}{9a + 1}$$

得到

$$b = \frac{1}{81a}$$

由

$$a + \frac{1}{9} + c + \frac{1}{9} + b + \frac{1}{3} = 1$$

可得

$$a + b + c = \frac{4}{9}$$

从而

$$a + \frac{1}{81a} + 3a = \frac{4}{9} \implies (18a - 1)^2 \implies a = \frac{1}{18}$$

故

$$a = \frac{1}{18}, \quad b = \frac{2}{9}, \quad c = \frac{1}{6}$$

□

Exercise 11.4

设 X 与 Y 是两个相互独立的随机变量, $X \sim U(0, 1)$, $Y \sim \text{Exp}(1)$. 试求:

1. X 与 Y 的联合密度函数;
2. $P(Y \leq X)$;
3. $P(X + Y \leq 1)$.

Proof.

$$p_X(x) = \chi_{[0,1]}(x)$$

$$p_Y(y) = \chi_{[0,\infty)}(y) e^{-y}$$

1. 由于 X, Y 独立,

$$p_{(X,Y)}(x,y) = p_X(x) p_Y(y) = \begin{cases} e^{-y}, & 0 \leq x \leq 1, y \geq 0 \\ 0, & \text{其它} \end{cases}$$

- 2.

$$P(Y \leq X) = \int_{-\infty}^{\infty} \int_{-\infty}^s p_{(X,Y)}(s,t) dt ds = \int_0^1 \int_0^s e^{-t} dt ds = e^{-1}$$

3.

$$\begin{aligned}
 P(X + Y \leq 1) &= P(Y \leq 1 - X) = \int_{-\infty}^{\infty} \int_{-\infty}^{1-s} p_{(X,Y)}(s, t) \, dt \, ds \\
 &= \int_0^1 \int_0^{1-s} e^{-t} \, dt \, ds \\
 &= e^{-1}
 \end{aligned}$$

□

Exercise 11.5

设随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} 1, & |x| < y, 0 < y < 1, \\ 0, & \text{其他.} \end{cases}$$

试求:

1. 边际密度函数 $p_X(x)$ 和 $p_Y(y)$;
2. X 与 Y 是否独立?

Proof. 1.

$$p_X(x) = \chi_{\{x \in (-1,1)\}} \int_{-\infty}^{\infty} p(x, y) \, dy = \int_{|x|}^1 1 \, dy = (1 - |x|) \chi_{\{x \in (-1,1)\}}$$

$$p_Y(y) = \int_{-\infty}^{\infty} p(x, y) \, dx = \chi_{[0,1]}(y) \int_{-y}^y 1 \, dx = \begin{cases} 2y, & y \in (0, 1) \\ 0, & \text{其它} \end{cases}$$

2.

$$p_X(x) p_Y(y) = \chi_{\{x \in (-1,1)\}} \chi_{\{y \in (0,1)\}} (1 - |x|) 2y$$

因此

$$p(x, y) \neq p_X(x) p_Y(y)$$

故 X, Y 不独立.

□

Exercise 11.6

设二维随机变量 (X, Y) 的联合密度函数如下, 试问 X 与 Y 是否相互独立?

1.

$$p(x, y) = \begin{cases} 24xy, & 0 < x < 1, 0 < y < 1, 0 < x + y < 1, \\ 0, & \text{其他;} \end{cases}$$

2.

$$p(x, y) = \begin{cases} 12xy(1-x), & 0 < x < 1, 0 < y < 1, \\ 0, & \text{其他;} \end{cases}$$

3.

$$p(x, y) = \begin{cases} \frac{21}{4}x^2y, & x^2 < y < 1, \\ 0, & \text{其他.} \end{cases}$$

Proof. 1.

$$p_X(x) = \chi_{\{x \in (0,1)\}} \int_0^{1-x} 24xy \, dy = \chi_{\{x \in (0,1)\}} 12x(1-x)^2$$

$$p_Y(y) = \chi_{\{y \in (0,1)\}} 12y(1-y)^2$$

于是

$$p_X(x) p_Y(y) = \chi_{\{x \in (0,1), y \in (0,1)\}} 12^2 xy(1-x)^2(1-y)^2$$

$$p_X(x) p_Y(y) \neq p(x, y)$$

故 X, Y 不独立.

2.

$$p(x, y) = 12x(1-x) \cdot y$$

是可分离变量的, 故 X, Y 独立.

3.

$$p_X(x) = \chi_{\{x \in (-1,1)\}} \int_{x^2}^1 \frac{21}{4}x^2y \, dy = \chi_{\{x \in (-1,1)\}} \frac{21}{8}x^2(1-x^4)$$

$$p_Y(y) = \chi_{\{y \in (0,1)\}} \int_{-\sqrt{y}}^{\sqrt{y}} \frac{21}{4}x^2y \, dx = \frac{7}{2}\chi_{\{y \in (0,1)\}}y^2\sqrt{y}$$

因此

$$p_X(x) \cdot p_Y(y) = \frac{147}{16} \chi_{\{x \in (-1,1), y \in (0,1)\}} x^2 y^2 (1-x^2) \sqrt{y}$$

$$p_X(x) \cdot p_Y(y) \neq p(x, y)$$

因此 X, Y 不独立.

□

Exercise 11.7

在长为 a 的线段的中点的两边随机地各选取一点, 求两点间的距离小于 $a/3$ 的概率.

Proof. 以中点为原点, 建立水平方向的坐标系, 令 X, Y 分别为选取两点的坐标. 则 $X \sim U\left(\left(-\frac{a}{2}, 0\right)\right), Y \sim U\left(\left(0, \frac{a}{2}\right)\right)$.

$$p_X(x) = \chi_{\{x \in (-\frac{a}{2}, 0)\}} \frac{2}{a}, \quad p_Y(y) = \chi_{\{y \in (0, \frac{a}{2})\}} \frac{2}{a}$$

由于 X, Y 独立,

$$p_{(X,Y)}(x, y) = \chi_{\{x \in (-\frac{a}{2}, 0), y \in (0, \frac{a}{2})\}} \frac{4}{a^2}$$

两点距离小于 $\frac{a}{3}$ 的概率为

$$\int \int_{\{Y-X \leq \frac{a}{3}\}} p_{(X,Y)}(x, y) \, dx \, dy = \int_{-\frac{a}{3}}^0 \int_0^{\frac{a}{3}+x} \frac{4}{a^2} \, dy \, dx = \frac{2}{9}$$

□

Exercise 11.8

设二维随机变量 (X, Y) 的联合分布列为

$X \backslash Y$	1	2	3
0	0.05	0.15	0.20
1	0.07	0.11	0.22
2	0.04	0.07	0.09

试分别求 $U = \max\{X, Y\}$ 和 $V = \min\{X, Y\}$ 的分布列.

Proof. 解

$U = \max\{X, Y\}$ 的可能取值为 1, 2, 3.

$$P(U = 1) = P(0, 1) + P(1, 1) = 0.05 + 0.07 = 0.12,$$

$$P(U = 2) = P(0, 2) + P(1, 2) + P(2, 1) + P(2, 2) = 0.15 + 0.11 + 0.04 + 0.07 = 0.37,$$

$$P(U = 3) = P(Y = 3) = 0.20 + 0.22 + 0.09 = 0.51.$$

$V = \min\{X, Y\}$ 的可能取值为 0, 1, 2.

$$P(V = 0) = P(X = 0) = 0.40,$$

$$P(V = 1) = P(1, 1) + P(1, 2) + P(1, 3) + P(2, 1) = 0.07 + 0.11 + 0.22 + 0.04 = 0.44,$$

$$P(V = 2) = P(2, 2) + P(2, 3) = 0.07 + 0.09 = 0.16.$$

故 U 和 V 的分布列分别为

U	1	2	3
P	0.12	0.37	0.51

V	0	1	2
P	0.40	0.44	0.16

□

Exercise 11.9

设 X 和 Y 是相互独立的随机变量, 且 $X \sim \text{Exp}(\lambda)$, $Y \sim \text{Exp}(\mu)$. 如果定义随机变量

$$Z = \begin{cases} 1, & X \leq Y, \\ 0, & X > Y, \end{cases}$$

求 Z 的分布列.

Proof.

$$\begin{aligned} P(Z = 1) &= P(X \leq Y) = \int_0^\infty \int_0^y \lambda e^{-\lambda x} \mu e^{-\mu y} dx dy = \int_0^\infty \mu e^{-\mu y} - \mu e^{-(\lambda+\mu)y} dy \\ &= 1 - \frac{\mu}{\lambda + \mu} = \frac{\lambda}{\lambda + \mu} \end{aligned}$$

$$P(Z = 0) = 1 - P(Z = 1) = \frac{\mu}{\lambda + \mu}$$



Exercise 11.10

设随机变量 X 和 Y 的分布列分别为

X	-1	0	1
P	1/4	1/2	1/4

Y	0	1
P	1/2	1/2

已知 $P(XY = 0) = 1$, 试求 $Z = \max\{X, Y\}$ 的分布列.

Solution. 由条件 $P(XY = 0) = 1$ 知 $P(X \neq 0, Y \neq 0) = 0$, 故

$$P(X = -1, Y = 1) = P(X = 1, Y = 1) = 0.$$

结合边缘分布列, 可得联合分布中概率非零的项为

$$P(X = -1, Y = 0) = P(X = -1) - P(X = -1, Y = 1) = 1/4,$$

$$P(X = 1, Y = 0) = P(X = 1) - P(X = 1, Y = 1) = 1/4,$$

$$P(X = 0, Y = 1) = P(Y = 1) - P(X = -1, Y = 1) - P(X = 1, Y = 1) = 1/2.$$

对于 $Z = \max\{X, Y\}$, 其可能取值为 0, 1.

$$P(Z = 0) = P(X = -1, Y = 0) = 1/4,$$

$$P(Z = 1) = P(X = 1, Y = 0) + P(X = 0, Y = 1) = 1/4 + 1/2 = 3/4.$$

故 Z 的分布列为

Z	0	1
P	1/4	3/4



Exercise 11.11

设随机变量 X, Y 独立同分布, 在以下情况下求随机变量 $Z = \max\{X, Y\}$ 的分布列.

1. X 服从 $p = 0.5$ 的 0-1 分布.
2. X 服从几何分布, 其中

$$P(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, \dots$$

Proof. 1.

$$P(Z = 0) = P(X = Y = 0) = P(X = 0)P(Y = 0) = 0.5 \times 0.5 = 0.25$$

$$P(Z = 1) = 1 - P(Z = 0) = 0.75$$

2.

$$P(X \leq k) = \sum_{i=1}^k (1-p)^{i-1} p = 1 - (1-p)^k$$

于是

$$\begin{aligned} P(Z \leq k) &= P(X \leq k, Y \leq k) = P(X \leq k)P(Y \leq k) = P(X \leq k)^2 \\ &= \left(1 - (1-p)^k\right)^2 \end{aligned}$$

$$\begin{aligned} P(Z = k) &= P(Z \leq k) - P(Z \leq k-1) \\ &= \left(1 - (1-p)^k\right)^2 - \left(1 - (1-p)^{k-1}\right)^2 \\ &= \left((1-p)^{k-1} - (1-p)^k\right) \left(2 - (1-p)^k - (1-p)^{k-1}\right) \\ &= (1-p)^{k-1} p \left(2 - (1-p)^k - (1-p)^{k-1}\right), \quad k = 1, 2, \dots \end{aligned}$$

□

Exercise 11.12

设 X 和 Y 为两个随机变量, 且

$$P(X \geq 0, Y \geq 0) = \frac{3}{7}, \quad P(X \geq 0) = P(Y \geq 0) = \frac{4}{7}.$$

试求 $P(\max\{X, Y\} \geq 0)$.

Solution. 记 $A = \{X \geq 0\}$, $B = \{Y \geq 0\}$. 注意 $\{\max\{X, Y\} \geq 0\}$ 等于 $A \cup B$.

$$\begin{aligned} P(\max\{X, Y\} \geq 0) &= P(A \cup B) \\ &= P(A) + P(B) - P(AB) \\ &= \frac{4}{7} + \frac{4}{7} - \frac{3}{7} \\ &= \frac{5}{7}. \end{aligned}$$

Exercise 11.13

设 X 与 Y 的联合密度函数为

$$p(x, y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0, \\ 0, & \text{其他.} \end{cases}$$

试求以下随机变量的密度函数:

1. $Z = (X + Y)/2$.
2. $Z = Y - X$.

Proof. 1. 由于

$$p(x, y) = (\chi_{\{x>0\}}e^{-x})(\chi_{\{y>0\}}e^{-y})$$

是可分离变量的, 故 X, Y 独立, 且有

$$p_X(x) = \chi_{\{x>0\}}e^{-x}, \quad p_Y(y) = \chi_{\{y>0\}}e^{-y}$$

令 $W = 2Z$, 则

$$\begin{aligned} p_W(t) &= p_X * p_Y(t) = \int p_X(t-x)p_Y(x) dx \\ &= \int \chi_{\{x<t\}}e^{x-t}\chi_{\{x>0\}}e^{-x} dx \\ &= \chi_{\{t>0\}} \int_0^t e^{-t} dx \\ &= \chi_{\{t>0\}}te^{-t} \end{aligned}$$

从而

$$p_Z(z) = 2p_W(2z) = 2\chi_{\{z>0\}}2ze^{-2z}$$

2.

$$p_{-X}(x) = p_X(-x) = \chi_{\{x<0\}}e^x$$

$$\begin{aligned}
p_Z(z) &= p_Y * p_{-X}(z) = \int p_Y(z-x) p_{-X}(x) dx \\
&= \int \chi_{\{x < z\}} e^{x-z} (\chi_{\{x < 0\}} e^x) dx \\
&= \begin{cases} \int_{-\infty}^0 e^{2x-z} dx, & z \geq 0 \\ \int_{-\infty}^z e^{2x-z} dx, & z < 0 \end{cases} \\
&= \begin{cases} \frac{1}{2} e^{-z}, & z \geq 0 \\ \frac{1}{2} e^z, & z < 0 \end{cases} \\
&= \frac{1}{2} e^{-|z|}
\end{aligned}$$

□

Exercise 11.14

设随机变量 X 与 Y 相互独立, 试在以下情况下求 $Z = X + Y$ 的密度函数:

1. $X \sim U(0, 1), Y \sim U(0, 1)$
2. $X \sim U(0, 1), Y \sim \text{Exp}(1)$

Proof. 1.

$$\begin{aligned}
p_Z(z) &= p_X * p_Y(z) = \int p_X(z-x) p_Y(x) dx \\
&= \int \chi_{\{0 \leq z-x \leq 1\}} \chi_{\{0 \leq x \leq 1\}} dx \\
&= \int \chi_{\{z-1 \leq x \leq z\}} \chi_{\{0 \leq x \leq 1\}} dx \\
&= \chi_{\{0 \leq z \leq 1\}} z + \chi_{\{1 < z \leq 2\}} (2-z) \\
&= \begin{cases} z, & z \leq 1 \\ (2-z), & 1 < z \leq 2 \\ 0, & \text{其它} \end{cases}
\end{aligned}$$

2.

$$\begin{aligned}
p_Z(z) &= p_X * p_Y(z) = \int p_X(z-x) p_Y(x) dx \\
&= \int \chi_{\{0 \leq z-x \leq 1\}} \chi_{\{x \geq 0\}} e^{-x} dx \\
&= \int \chi_{\{z-1 \leq x \leq z\}} \chi_{\{x \geq 0\}} e^{-x} dx \\
&= \chi_{\{0 \leq z < 1\}} \int_0^z e^{-x} dx + \chi_{\{z \geq 1\}} \int_{z-1}^z e^{-x} dx \\
&= \begin{cases} 0, & z < 0 \\ 1 - e^{-z}, & 0 \leq z < 1 \\ (e-1)e^{-z}, & z \geq 1 \end{cases}
\end{aligned}$$

□

Exercise 11.15

设某一设备装有 3 个同类的电器元件, 元件工作相互独立, 且工作时间都服从参数为 λ 的指数分布. 当 3 个元件都正常工作时, 设备才正常工作. 试求设备正常工作时间 T 的密度函数.

Proof. 记三个设备的工作时间分别为 X, Y, Z . 则

$$T = \min\{X, Y, Z\}$$

$$\{T \geq t\} = \{X \geq t, Y \geq t, Z \geq t\}$$

由独立性

$$P(T \geq t) = P(X \geq t)P(Y \geq t)P(Z \geq t)$$

由于它们都服从参数为 λ 的指数分布,

$$P(X \geq t) = P(Y \geq t) = P(Z \geq t) = \int_t^\infty \lambda e^{-\lambda x} dx = \chi_{\{t \geq 0\}} e^{-\lambda t}$$

故

$$P(T \geq t) = \chi_{\{t \geq 0\}} e^{-3\lambda t}$$

从而 $P(T \leq t) = 1 - P(T \geq t) = 1 - e^{-3\lambda t}$, 进而

$$p_T(t) = \frac{d}{dt}P(T \leq t) = 3\lambda \chi_{\{t \geq 0\}} e^{-3\lambda t}$$

□

Exercise 11.16

设随机变量 $X_1, X_2, \dots, X_n$ 相互独立, 且 $X_i \sim \text{Exp}(\lambda_i)$, 试证:

$$P(X_i = \min\{X_1, X_2, \dots, X_n\}) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}.$$

Proof. 由于 $X_1, \dots, X_n$ 相互独立,

$$p_{(X_1, \dots, X_n)}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n) = \lambda_1 e^{-\lambda_1 x_1} \cdots \lambda_n e^{-\lambda_n x_n}$$

注意到

$$P(X_i = \min\{X_1, X_2, \dots, X_n\}) = P(X_i \leq X_1, X_i \leq X_2, \dots, X_i \leq X_n)$$

其中

$$\begin{aligned} & P(X_i \leq X_1, \dots, X_i \leq X_n) \\ &= \int_0^\infty \int_{x_i}^\infty \cdots \int_{x_i}^\infty \lambda_1 e^{-\lambda_1 x_1} \cdots \lambda_n e^{-\lambda_n x_n} dx_1 \cdots dx_{i-1} \cdots dx_{i+1} \cdots dx_n dx_i \\ &= \int_0^\infty \int_{x_i}^\infty \cdots \int_{x_i}^\infty e^{-\lambda_1 x_i} \lambda_2 e^{-\lambda_2 x_2} \cdots \lambda_n e^{-\lambda_n x_n} dx_2 \cdots dx_{i-1} \cdots dx_{i+1} \cdots dx_n dx_i \\ &\dots \\ &= \lambda_i \int_0^\infty \left(\prod_{j \neq i} e^{-\lambda_j x_i} \right) \lambda_i e^{-\lambda_i x_i} dx_i \\ &= \lambda_i \int_0^\infty e^{-(\lambda_1 + \dots + \lambda_n)x_i} dx_i \\ &= \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n} \end{aligned}$$

因此

$$P(X_i = \min\{X_1, X_2, \dots, X_n\}) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}$$

□

Exercise 11.17

设连续随机变量 $X_1, X_2, \dots, X_n$ 独立同分布, 试证:

$$P(X_n > \max\{X_1, X_2, \dots, X_{n-1}\}) = \frac{1}{n}.$$

Proof. 由于 $X_1, \dots, X_n$ 相互独立,

$$p_{(X_1, \dots, X_n)}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$$

由于 $X_1, X_2, \dots, X_n$ 同分布, 令 $p(x) = p_{X_n}(x)$, 则 $p_{X_i}(x) = p(x), i = 1, \dots, n$. 于是

$$p_{(X_1, \dots, X_n)}(x_1, \dots, x_n) = p(x_1) \cdots p(x_n)$$

$$\begin{aligned} P(X_n > \max\{X_1, \dots, X_{n-1}\}) \\ &= P(X_n > X_1, \dots, X_n > X_{n-1}) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_n} p(x_1) \cdots p(x_n) dx_1 \cdots dx_{n-1} dx_n \end{aligned}$$

记 $F(x) = \int_{-\infty}^x p(x) dx$, 则 $F'(x) = p(x)$, 且

$$\begin{aligned} &\int_{-\infty}^{\infty} \int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_n} p(x_1) \cdots p(x_n) dx_1 \cdots dx_{n-1} dx_n \\ &= \int_{-\infty}^{\infty} (F(x_n))^{n-1} p(x_n) dx_n \\ &= \int_{-\infty}^{\infty} (F(x_n))^{n-1} dF(x_n) \\ &= \left[\frac{1}{n} (F(x_n))^n \right]_{-\infty}^{\infty} \\ &= \frac{1}{n} (1 - 0) = \frac{1}{n} \end{aligned}$$

□